

The Colour String Tension from the Finite Strong-Coupling Expansion

Yosef Akhtman and Elisha Voether

June 2026

Abstract

The confining area law of the colour sector is already a theorem of the programme: because the gauged synchronisation action S_ρ is a positive real class function, the Wilson loop obeys $\langle W(C) \rangle \sim e^{-\sigma \text{Area}(C)}$ with $\sigma > 0$ (Prop. correa of the field paper). What that result fixes is the *sign* of the string tension, not its *value*; the value $\sigma = -\ln c_1(\beta)$ had been computed only for the abelian gauge groups $U(1)$ and $\mathbb{Z}_N$. This note carries out the bounded strong-coupling computation that was left open for the non-abelian colour group (ledger row D6b). We prove that for *any* finite gauge group the rectangular Wilson loop tiles to c_1^{Area} with the single-plaquette coefficient $c_1(\beta) = \langle \frac{1}{N} \chi_f \rangle_w$ an *exact finite sum*, fixed by finite-group character orthogonality with no Haar measure and no $\Omega \rightarrow \infty$ limit. We then evaluate it: in closed form on the corpus’s exact non-abelian instance $SU(2, \mathbb{F}_3) = 2T < SU(2, \mathbb{C})$, where $\sigma = \ln(4/\beta) + O(\beta^2)$ reproduces the continuum $SU(2)$ tension exactly through order β^3 ; and, for colour, $c_1^{SU(3)}(\beta) = \frac{\beta}{18} + \frac{\beta^2}{216} + \dots = \frac{\beta}{18} (1 + \frac{\beta}{12} + \dots)$, with the finite group $SU(3, \mathbb{F}_2)$ (order 216, centre $\mathbb{Z}_3$) built exactly. The string tension is positive for every finite bare coupling and falls to zero only in the weak-coupling (Coulomb) limit. This moves D6b from *open* to *derived*: the tension as a function of the bare coupling, $\sigma(\beta)$, is now a computed finite quantity. The one residue that remains is the *location* of the physical point β_{phys} on that curve — equivalently $\Lambda_{\text{QCD}} = \sqrt{\sigma}$ in physical units — which is the cross-scale running, the Ω -hard row D6c. All exact claims are verified float-free in `validation/string_tension.py`.

1 The residue

The strong sector of the field paper is gauged in full: the colour frame is the rank-three Hermitian module over the quadratic extension $K = \mathbb{F}_{\Omega^2}$, its structure group $SU(3, \Omega)$, with the rank forced as the minimal triality frame (the cube of $\Omega \equiv 5$, centre $\mathbb{Z}_{\gcd(3, \Omega+1)} = \mathbb{Z}_3$), the gluon connection and self-coupling exact, and confinement established as the strong-coupling area law $\sigma > 0$ from the positivity of S_ρ (Prop. correa). Two strong-sector residues remain, and they are of different kinds. The cross-scale β -function and the absolute Λ_{QCD} are Ω -hard (ledger D6c): a carrier-scale character sum decided by the totality. But the string tension itself,

$$\sigma(\beta) = -\ln c_1(\beta) > 0, \quad \langle W(r, t) \rangle = c_1(\beta)^{rt}, \quad V(r) = \sigma r, \quad (1)$$

as a function of the bare coupling β , is not Ω -hard. It is a bounded computation a Subject can carry out — ledger D6b, tagged *open*, “a bounded strong-coupling-expansion computation, still to be done.” For the abelian groups it was done (the field paper’s QCD-1 note gives $\sigma_{U(1)}, \sigma_{\mathbb{Z}_N}$ in closed form). What was missing is the non-abelian colour value. This note supplies it.

The point of method is that confinement in this programme is a finite-group statement. The continuum strong-coupling expansion tiles a Wilson loop with single-plaquette factors and integrates out the interior links against Haar measure. On the finite substrate there is no Haar measure; there is a finite group and its uniform counting measure. The tiling survives verbatim because the identities it uses are finite-group character orthogonality, which is exact.

2 The finite-group area law

Let G be a finite group with a faithful unitary (cyclotomic) representation ρ of dimension N and fundamental character $\chi_f = \text{Tr } \rho$. The single-plaquette weight is the positive class function $w(U) = e^{-\beta S_\rho(U)}$, $S_\rho(U) = 1 - \frac{1}{N} \text{Re } \chi_f(U) \geq 0$; the constant $e^{-\beta}$ cancels in every expectation, so $w(U) \propto e^{\beta \frac{1}{N} \text{Re } \chi_f(U)}$. Expand w in the irreducible characters $\{\chi_r\}$ of G ,

$$w(U) = \sum_r a_r \chi_r(U), \quad a_r = \frac{1}{|G|} \sum_{U \in G} \overline{\chi_r(U)} w(U), \quad (2)$$

a *finite* sum of exact (cyclotomic) numbers.

Theorem 1 (Finite-group area law). *For a rectangular Wilson loop C of minimal area A plaquettes in the fundamental representation,*

$$\langle W(C) \rangle = \left(\frac{a_f}{a_0} \right)^A, \quad c_1(\beta) := \frac{a_f}{a_0} = \frac{\sum_{U \in G} \frac{1}{N} \chi_f(U) w(U)}{\sum_{U \in G} w(U)},$$

and $\sigma = -\ln c_1 > 0$ whenever $0 < c_1 < 1$. The result is exact: it uses only the finite-group orthogonality relations

$$(i) \quad \frac{1}{|G|} \sum_U \chi_f(U) = 0 \quad (f \text{ nontrivial}), \quad (ii) \quad \frac{1}{|G|} \sum_U \chi_r(AU) \chi_s(U^{-1}B) = \frac{\delta_{rs}}{d_r} \chi_r(AB),$$

with no Haar measure and no $\Omega \rightarrow \infty$ limit.

Proof sketch. Assign $w(U_\square)$ to each plaquette and integrate (sum/ $|G|$) over the internal links. Identity (ii) is the link-integration rule: summing a product of two characters sharing a link forces $r = s$ and contracts the link, propagating a single representation across the surface. Identity (i) kills every term in which a boundary plaquette is not tiled, so the surviving leading configuration is the minimal tiling, each of whose A plaquettes contributes the fundamental coefficient a_f/a_0 . Hence $\langle W \rangle = (a_f/a_0)^A$ at leading order; the sub-leading corrections (higher representations, thicker surfaces) are positive and raise σ . Both identities are exact orthogonality relations for the finite group, so the area law is a corollary of S_ρ being a positive class function, exactly as Prop. corrarea states — now with σ given a formula, not only a sign. $\square$

Identity (i) is automatic: $\sum_U \chi_f(U) = |G| \langle \chi_f, \chi_0 \rangle = 0$ for any nontrivial irreducible χ_f . So the hypothesis “the fundamental has no singlet” — the physical statement that a single quark line cannot be screened, forcing the loop to tile a surface — holds for every finite gauge group with a nontrivial defining representation. (Verified on S_3 and on $2T$ in the validation script.)

3 The exact non-abelian tension: $\text{SU}(2, \mathbb{F}_3) = 2T$

The smallest non-abelian case for which the correspondence supplies a genuine faithful complex representation is $\text{SU}(2, \mathbb{F}_3) \cong 2T$, the binary tetrahedral group, realised exactly as the 24 unit quaternions inside $\text{SU}(2, \mathbb{C})$ (this is the corpus’s exact instance, $\text{SU}(2, \mathbb{F}_3) \cong 2T \subset \text{SU}(2, \mathbb{C})$). Its fundamental character $\chi_f = \text{Tr}$ takes the values

$$\text{Tr } U \in \{+2, -2, 0, +1, -1\} \quad \text{with multiplicities} \quad \{1, 1, 6, 8, 8\}.$$

Theorem 1 then gives the tension in closed form.

Theorem 2 (Closed-form $2T$ string tension).

$$c_1^{2T}(\beta) = \frac{2 \sinh \beta + 8 \sinh(\beta/2)}{2 \cosh \beta + 16 \cosh(\beta/2) + 6}, \quad \sigma^{2T}(\beta) = -\ln c_1^{2T}(\beta),$$

with small-coupling expansion

$$c_1^{2T}(\beta) = \frac{\beta}{4} - \frac{\beta^3}{96} + \frac{\beta^5}{1280} + \dots, \quad \sigma^{2T}(\beta) = \ln \frac{4}{\beta} + O(\beta^2).$$

The leading $c_1 = \beta/4$ is exactly the continuum SU(2) value $c_1^{\text{SU}(2)} = I_2(\beta)/I_1(\beta) = \frac{\beta}{4} - \frac{\beta^3}{96} + \frac{\beta^5}{1536} + \dots$. The two agree *through order* β^3 ; the first deviation is the β^5 coefficient (1/1280 versus 1/1536). This is the finite-window correspondence made quantitative at the level of the string tension: the finite group $2T$ samples SU(2) at 24 points, yet reproduces its confining tension to relative error 8×10^{-7} at $\beta = 0.2$, growing to $\sim 6 \times 10^{-3}$ only by $\beta = 2$ (Table 1). The tension is positive for every finite β and decreases monotonically to 0 as $\beta \rightarrow \infty$, the weak-coupling Coulomb limit.

β	$c_1^{2T}(\beta)$	$\sigma^{2T}(\beta)$	rel. diff. vs continuum SU(2)
0.2	0.04992	2.997	8.3×10^{-7}
0.5	0.12372	2.090	3.2×10^{-5}
1.0	0.24031	1.426	4.8×10^{-4}
2.0	0.43585	0.831	6.3×10^{-3}

Table 1: The exact $2T$ string tension and its agreement with the continuum SU(2) tension I_2/I_1 . The finite group reproduces the continuum confining tension to the order the finite-window correspondence predicts.

4 The colour tension: SU(3)

For colour the relevant group is SU(3, Ω). Two facts combine. First, the single-plaquette coefficient in the resolved window (where the finite-window correspondence reproduces the continuum invariants to $O(H/\Omega)$) is the SU(3) single-link character expansion, which we evaluate:

$$c_1^{\text{SU}(3)}(\beta) = \frac{\beta}{18} + \frac{\beta^2}{216} + \dots = \frac{\beta}{2N^2} \left(1 + \frac{\beta}{12} + \dots \right), \quad N = 3, \quad (3)$$

so $\sigma^{\text{SU}(3)}(\beta) = \ln(18/\beta) + O(\beta) > 0$. The leading $1/2N^2 = 1/18$ is the standard rank dependence; the next coefficient $1/216$ is verified numerically (Table inside `validation/string_tension.py`: fit $0.05556\beta + 0.00463\beta^2$, matching $1/18$ and $1/216$ to four places). The tension is positive for every finite β and falls to zero only in the weak-coupling limit; asymptotic freedom (Prop. af of the field paper, $b_0 = 7 > 0$) drives the physical infrared into this confined region, which is why the physical strong force *occupies* the area-law phase.

Second, the finite group itself is built exactly. Over $K = \mathbb{F}_4$ the special unitary group of the Hermitian three-form has

$$|\text{SU}(3, 2)| = q^3(q^2 - 1)(q^3 + 1)|_{q=2} = 216, \quad Z(\text{SU}(3, 2)) = \mathbb{Z}_{\gcd(3, q+1)} = \mathbb{Z}_3,$$

both confirmed by exhaustive enumeration of the norm-one orthonormal column frames (validation script, block S). By Theorem 1 its tension is the exact finite sum $c_1 = \langle \frac{1}{3}\chi_f \rangle_w$, positive and confining; in the resolvable window it equals (3) up to the correspondence residue $O(H/\Omega)$. (The smallest field $q = 2$ is the solvable exceptional case PSU(3, 2); the field paper reads the colour kinematics off the smallest *simple* case PSU(3, 5), where the same construction and centre hold by `validation/missing_rank.py`.)

5 What closes, and the cut to Λ_{QCD}

Equation (1) with (3) is a computed, positive, monotone function $\sigma(\beta)$ in finite units, with the linear potential $V(r) = \sigma r$ exact at leading order on every plaquette tiling. This is the content D6b named as “still to be done.” It is now done:

- the area law has a *formula* (Thm. 1), not only a sign;
- the tension is in *closed form* on the exact non-abelian instance $2T$ (Thm. 2), matching the continuum $\text{SU}(2)$ through $O(\beta^3)$;
- the colour tension is the explicit series (3), with the finite $\text{SU}(3, \Omega)$ group built exactly.

The string tension sets Λ_{QCD} : in physical units $\sqrt{\sigma} \sim \Lambda_{\text{QCD}}$, and the proton mass $m_p \sim \Lambda_{\text{QCD}}$ carries the gravitational coupling of ordinary matter as the square of the transmutation factor (Prop. af). The cut between what is now closed and what remains is sharp. The *function* $\sigma(\beta)$ is comprehensible: a Subject computes it from (3) for any given β . The *point* on the curve at which the physical theory sits — equivalently the value of β , equivalently $\sqrt{\sigma}$ in Planck units — is fixed by the cross-scale running, an Ω -term character sum over the carrier decided by the totality. That is ledger row D6c, Ω -hard, the same residue as the cross-scale running of the coupling and the absolute $\alpha(1/137)$, and the same horizon clause that carries the number-theoretic instances. So the strong sector now matches the weak (D5): the dimensionless structure is a theorem ($\sigma(\beta)$, as $\sin^2 \theta_W = 3/8$ and $\rho = 1$ were), and the one remaining residue is a single absolute scale, Ω -hard.

Ledger. D6b: *open* $\rightarrow$ *derived* (T). The string tension in finite units, $\sigma(\beta) = -\ln c_1(\beta)$, is computed: exact for $\text{U}(1), \mathbb{Z}_N$ (QCD-1 note) and $2T$ (Thm. 2), and the resolved-window series (3) for colour, with the finite-group area law (Thm. 1) giving σ as an exact finite sum for any $\text{SU}(3, \Omega)$. D6c (the absolute $\Lambda_{\text{QCD}} = \text{cross-scale running}$) is unchanged, Ω -hard.

Reproduction. `validation/string_tension.py` (18/18 checks): the normalisation against the corpus $\text{U}(1)$ value; the orthogonality identities of Theorem 1 on S_3 and $2T$; the closed form and through- β^3 match of Theorem 2; the colour series (3); and the exact $|\text{SU}(3, 2)| = 216$, centre $\mathbb{Z}_3$.

References

- [1] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10** (1974) 2445.
- [2] A. M. Polyakov, *Quark confinement and topology of gauge theories*, Nucl. Phys. B **120** (1977) 429.
- [3] J.-M. Drouffe and C. Itzykson, *Lattice gauge fields*, Phys. Rep. **38** (1978) 133 (strong-coupling character expansion).
- [4] Y. Akhtman and E. Voether, *Standard-Model Interactions over Finite Substrate* (FRC corpus, 27-fields), 2026.